Journal of Cleaner Production 378 (2022) 134593

Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Modeling carbon emission estimation for hinterland-based container intermodal network

Xiaoyan Guo [a], Junliang He [a,b,*], Mo Lan [d], Hang Yu [b], Wei Yan [b,c]

[a] Institutes of Logistics Science and Engineering, Shanghai Maritime University, Shanghai, China
[b] China Institute of FTZ Supply Chain, Shanghai Maritime University, Shanghai, PR China
[c] Engineering Research Center of Container Supply Chain Technology, Ministry of Education, Shanghai Maritime University, Shanghai, PR China
[d] College of Transport and Communications, Shanghai Maritime University, Shanghai, China

ARTICLE INFO

Handling Editor: Dr. Govindan Kannan

Keywords:
Hinterland-based container intermodal networks
Modeling carbon emission estimation
Carbon emissions composition
Carbon emission reduction driving factors
Carbon peaking

ABSTRACT

In order to deal with the increasing environmental challenges, the high carbon-emitting transport system keeps transitioning to achieve carbon peaking and carbon neutrality goals. This paper addresses the calculation of carbon emission estimation and identifying carbon reduction driving factors for the hinterland-based container intermodal network. Firstly, an analysis of the carbon emissions composition of the hinterland-based container intermodal networks is conducted. Furthermore, a carbon emission estimation model for the hinterland-based container intermodal network is developed. Subsequently, taking the Shanghai port and the Yangtze River Delta (YRD) hinterland as a case, we estimate the carbon emission of the hinterland-based container intermodal network under the Well-to-Wheel (WTW) and the Tank-to-Wheel (TTW) modes in the past ten years. Finally, the critical driving factors are identified by sensitivity analysis, and the variation of carbon emissions under different scenarios with carbon reduction policies are analyzed. The results indicate that the transportation structure and the fuel choices affect network emissions reductions significantly. In the current policy-based scenario, the WTW mode achieves the carbon peaking between 2030 and 2035, while the TTW mode peaks earlier.

1. Introduction

Under the pressure of the simultaneous growth of global energy demand and carbon emissions, transportation with high carbon-emitting has become an essential field for carbon reduction (Cui et al., 2022; Miklautsch and Woschank, 2022). As the second largest carbon-emitting sector in the world, transportation accounts for 24.46% of global carbon emissions in 2019, while transportation accounts for 9.17% of carbon emissions in China (IEA, 2021). Furthermore, road transport emissions in China account for about 72.3% of total transport emissions in 2021. Therefore, on one hand transport energy and technology improvement for emission reduction are committed to low-carbon transport (Yazan S. M. Altarazi et al., 2022; Gustafsson et al., 2021; Hoang et al., 2022). On the other hand, optimizing the transport modes also provides a new direction for transport sector emission reduction (de Miranda Pinto et al., 2018; Ramalho and Santos, 2021; Tao and Wu, 2021). The development of ports mainly relies on mature intermodal transport networks to achieve multi-route connections between ports and inland hinterlands (Zhang et al., 2013). The carbon emissions by the hinterland-based container intermodal network also have high effects on the decarbonization of the regions (Tao and Wu, 2021). In this context, the Chinese national has launched a series of policies to put in place the realization of a green, low-carbon and circular economic system, many of which give specific targets for the low-carbon development of the transport sector. For example, *The Action Plan for Reaching the Carbon Peak by 2030* proposed that the proportion of new energy vehicles would reach 40% by 2030, carbon emission intensity would drop by about 9.5% compared to 2020, and oil consumption in land transport would strive to reach the peak by 2030 (The State Council of the PRC, 2021a). *The* 14th *Five-Year Plan* proposed an average annual increase of more than 15% in Rail-Water container intermodal transport (The State Council of the PRC, 2021b). Therefore, there are two critical questions to discuss based on these policies: firstly, what are the impacts of these policies on carbon emission reductions in the transport sector? Second, based on a large number of transport participants and the difficulty of sharing information, what is the best way to estimate carbon emissions? Considering the above issues, we focus on a hinterland-based multi-modal transport network and study the carbon emission estimation of

* Corresponding author. 1550 Haigang Avenue, Institutes of Logistics Science and Engineering, Shanghai Maritime University, Shanghai, 201306, PR China.
E-mail address: jlhe@shmtu.edu.cn (J. He).

https://doi.org/10.1016/j.jclepro.2022.134593
Received 3 August 2022; Received in revised form 24 September 2022; Accepted 5 October 2022
Available online 10 October 2022
0959-6526/© 2022 Elsevier Ltd. All rights reserved.

multimode transport network.

Considering the above issues, we focus on a hinterland-based intermodal network and develops a carbon emission estimation model and a freight flow estimation model for the hinterland-based multimodal network. Taking the intermodal transport network of Shanghai port and the hinterland cities in the Yangtze River Delta region (YRD) as a case, we estimate the carbon emission for the network in the last ten years. Through the driving factors discussed by sensitivity analysis, we discuss the network's carbon emission and peek into various scenarios based on the existing policy targets. Overall, this study provided a practical, theoretical instrument for carbon emission evaluation in intermodal networks and a method for verifying the impact of policy decisions in the transport sector.

The paper structure of the rest is as follows: Section 2 introduces the intermodal carbon emissions findings and identifies research gaps. Section 3 presents the structure of the hinterland intermodal transport networks and the methodology for modeling carbon emissions. After that, in Section 4, we introduce the carbon emissions model for intermodal networks. In Section 5, taking the Shanghai port-YRD hinterland transportation system as a case, we estimate and discuss the network carbon emissions, the critical drivers, and the peak status of different scenarios. Section 6 summarizes this paper's contribution and management insights.

2. Literature review

The hinterland is the landward side of the seaport. And the seaport cargo hinterland is the cargo attraction area which is formed by the traffic flow of the seaport connected to the inland cargo origin or destination (Talley and Ng, 2018). Hinterland container intermodal transport is the significant medium to link the traffic flow of the port-hinterland. (Dai and Yang, 2020; Macharis and Bontekoning, 2004). Based on this, the numerous sets of hinterland container intermodal chains constitute the hinterland container intermodal network (Ghane-Ezabadi and Vergara, 2016).

The research on low-carbon transport attracts many focuses and more scholars have considered carbon emission factor in research in the intermodal transport field (Fan et al., 2018; Kaack et al., 2018; Miklautsch and Woschank, 2022; L. Wang et al., 2020). It is common for intermodal carbon emission studies to consider emission constraints in the optimization objectives. For example, the impact of carbon emissions is considered in intermodal cooperative scheduling (W. Wang et al., 2020), route optimization (Sun et al., 2022), and network design (Bouchery and Fransoo, 2015; Zhang et al., 2013). The single emission constraint mentioned in those studies fails to capture the complex emission processes in multimodal transportation. Thus, some scholars have turned to specialized multimodal emissions studies. They presented the intermodal carbon emission process by building mathematical models (Craig et al., 2013; de Miranda Pinto et al., 2018; Heinold and Meisel, 2018). These researches have focused on proving the carbon emission reduction effectiveness of shifting from road to rail. Compared to the above studies, Lin (2019; Pizzol (2019), and Ramalho and Santos (2021) provided a more comprehensive carbon emission research for complex intermodal system with the addition of short sea or inland waterway transport. However, the above studies emphasize the intermodal carbon emissions during transportation and ignore the emissions at intermodal nodes. Fahdi et al. (2021), Lättilä et al. (2013), L. Wang et al. (2020), and Wiercx et al. (2019) studied the emissions of multimodal nodes in the transshipment process. While Tao and Wu (2021) extended the research on the intermodal chain to the whole intermodal process. Although the existing work has provided valuable results on intermodal carbon emissions, these researches are still limited to one or several intermodal chains. They have failed to extend them to a broader network level. In fact, intermodal transport is a complex system comprising a collaborative system of nodes, routes, and networks (Ghane-Ezabadi and Vergara, 2016). The perspective of network not only can present the overall carbon emission level of regional freight transportation comprehensively but also can evaluate the influence of emission reduction policies from a more practical approach.

Carbon emission estimation methods include the top-down approach which is based on transport vehicle activity and the bottom-up approach which is based on fuel consumption data (IPCC, 2006). The complex multimodal transport processes make statistics of fuel consumption more difficult. Therefore, we mentioned above literature mostly models emissions by the bottom-up approach. The impact of transport vehicles on climate change is often analyzed from the well-to-wheel (WTW) perspective (Gustafsson et al., 2021), consisting of two phases: well-to-tank (WTT) and tank-to-wheel (TTW) (Dreier et al., 2018). WTW is a life-cycle evaluation mode developed for transport fuels and energy carriers that cover resource extraction to vehicle operation, excluding the vehicle's life cycle (Gustafsson et al., 2021). In the TTW mode, electrical energy is considered to be zero carbon emission. In the existing studies on intermodal emissions, there are few works that distinguish between the two emission modes. ECOTRANSIT (2019) provided a detailed description of two multimodal carbon emission estimation modes. Tao and Wu (2021) established a generalized carbon emission analysis framework based on the revised ASIF (Activity - Structure - Intensity-Factor) model that considers the whole process of intermodal transport on the WTW emission. However, the revised ASIF method fails to present the process of carbon emission generation in intermodal transport networks intuitively. So, a model based on the emission generation process is needed to track the carbon emission process of multimodal transport networks and to distinguish between WTW and TTW modes.

To this end, this paper fills those gaps and develops a detailed description and modeling of the carbon emissions of the whole process from the perspective of regional hinterland-based intermodal transport networks. Furthermore, the study provides a freight flow estimation model for the carbon emission on intermodal networks. Finally, taking the intermodal transport network of Shanghai port and the hinterland cities in the Yangtze River Delta region (YRD) as a case, we estimate the carbon emission for the network in the last ten years on the WTW mode and the TTW mode.

3. Analysis of carbon emissions composition of the hinterland-based container intermodal networks

3.1. Structure of the hinterland-based container intermodal networks

We describe the hinterland-based container intermodal network as building on the transport network among ports and hinterlands. And this network depends on the freight flow in different periods. Within the network, containers move along the intermodal chain between ports and destinations through intermodal nodes and vehicles, as shown in Fig. 1. To reduce the complexity of the network, we make the following definitions and assumptions about the network scope of this paper.

(1) The intermodal transport network in this study is based on the freighted flows between nodes. The network estimation model of carbon emission evaluates the carbon emission in a specific period. Any node in the network can rely on its freight flow as an origin point.
(2) The hinterland transportation of containers includes roads, inland waterways and railways. The last mile distribution depends on road transport.

3.2. Carbon emission estimation for the intermodal networks

3.2.1. The framework of the estimation methodology

Based on the hinterland-based intermodal network mentioned in section 2.1, we develop a network carbon emission model from the port container flows. As shown in Fig. 2, firstly, the carbon emission

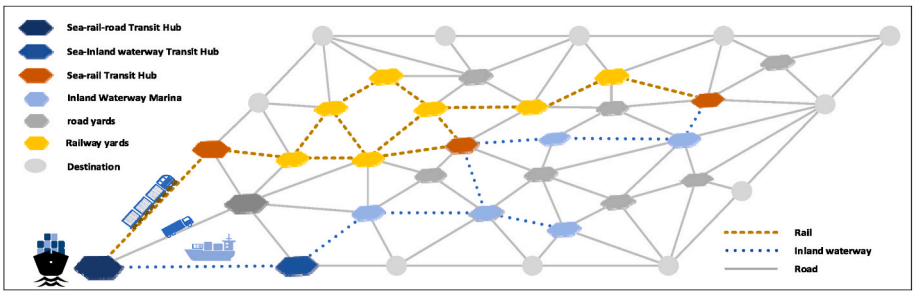


Fig. 1. Hinterland-based multimodal transport network.

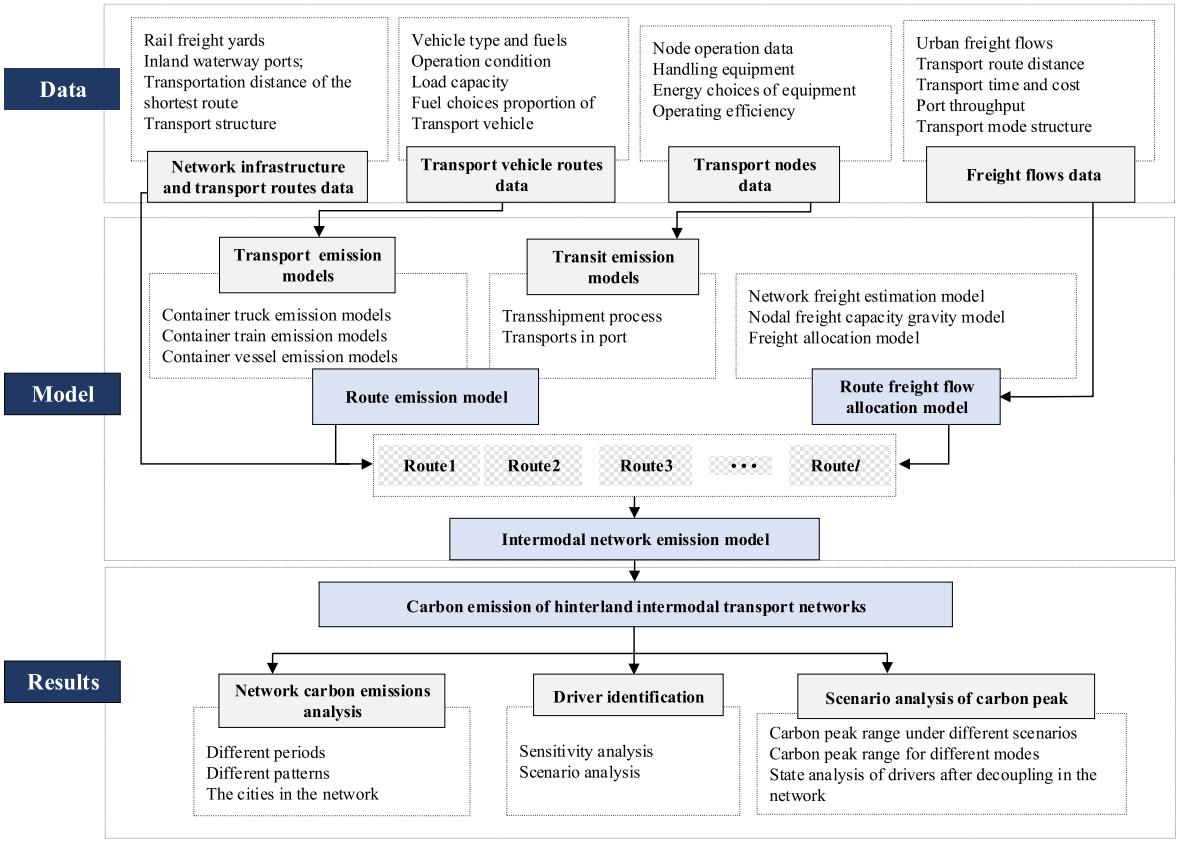


Fig. 2. Methodological framework.

estimation models of the intermodal network develop in two stages, including the transport and transfer. Secondly, we present a freight flows model to obtain the container freight flows from the port to nodes. Then combined flows with the carbon emission estimation model to obtain the carbon emission of the network. Similarly, this model can be used to estimate carbon emissions in a regional intermodal network by using each node in the region as a flow sending point and repeating the above calculation steps.

3.2.2. *The estimated emission in the model*

The carbon emissions of the transport process depend on fuel consumption and carbon emission factors. The fuel consumption depends

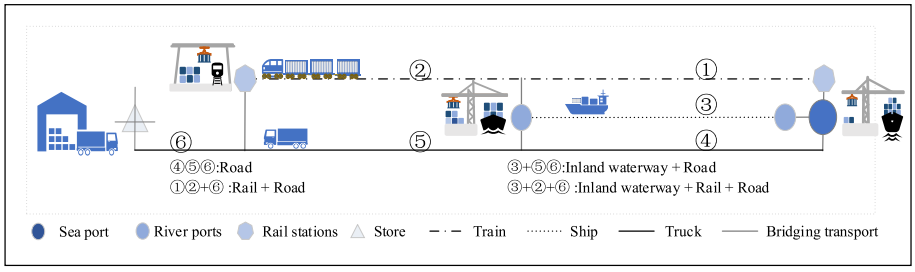


Fig. 3. Container transport process in intermodal transport.

on the particular characteristics and physical conditions of the transport vehicle (Jiang et al., 2020; Li and Zhang, 2020), including the specific vehicle type, capacity utilization, and driving conditions. Therefore, this research analyzes the energy consumption of container transport on road, rail, and ship, as shown in Fig. 3, though considering the factors of load weight, capacity utilization, and road conditions.

The energy consumption of container trucking is more significantly affected by the load condition (Lin, 2019), which includes the actual loading (Gustafsson et al., 2021) and empty miles. While road conditions also lead to the difference in energy consumption between highways and urban roads (Ferreira et al., 2020). A quarter of the energy consumption of container trucks is diesel (Hao et al., 2012). LNG and electricity are gradually being applied, as well as biodiesel (Dreier et al., 2018; Zhou et al., 2017). Therefore, the truck energy model in this paper focused on loading and considered the impact of different energy sources and road conditions on carbon emissions.

The Railway transport process in our study includes the departure of trains at the marshaling station until arrival at the destination before the break-up. Railway energy consumption varies in different transport sections (Chocholac et al., 2021). To reduce calculations complexity, we introduce the energy consumption factor per ton-kilometer in the operating section for diesel and electric locomotives. This factor consists of the gross weight of the locomotive, container wagon and freight. To make the model represent the relationship between freight weight and energy consumption, we converted the gross energy consumption into net by the coefficient of the gross and the net train weight (ECO-TRANSIT, 2019).

The ship energy consumption estimation should proceed from the ship characteristics and operating conditions. According to previous studies on energy consumption and carbon emissions of ship transportation (Cariou et al., 2019; Sun et al., 2013; Xiao et al., 2022), ship speed is the most sensitive factor, followed by capacity utilization, both of which are embodied in the engine load (IMO, 2020). Water and air resistance also have a significant influence on the ship's engine load. Thus, the model introduces a resistance coefficient (IMO, 2014).

The carbon emission estimation of the multimodal transfer process should consider the energy consumption of each operation period (Fahdi et al., 2021). We integrate multimodal nodes' operation and efficiency (He et al., 2021; Wiercx et al., 2019). On the one hand, the container transshipment operations include loading and unloading operations and horizontal transport operations, which vary depending on the different terminals, as shown in Fig. 4. Trucks can usually transfer in port directly. While transshipment of trains and ships depends on the infrastructure of port to consider whether it is possible to transfer directly in port, or else they have to be connected by trucks for short connections. On the other hand, we also consider the energy consumption of transport vehicles during assisting container handlings, such as ships in port, transport vehicles waiting in line, and locomotive traction.

4. Modeling the carbon emission estimation

The carbon emission estimation model includes three parts: the total network model, the transport process model and the transshipment process model. The notations in the model are shown in Table 1.

4.1. Total carbon emission estimation model of the intermodal transport network

The carbon emission of the intermodal network $E_h^{network}$ for hub node h is as follows:

$$E_h^{network} = \sum_{l=1}^{L} E_h^l, \forall h \in I \tag{4–1}$$

$$\sum_{l=1}^{L} Q_{l,p}^{network} \cdot X_{hd}^l = Q_{h,p}^{network}, \forall h,d \in I; \forall p \in P \tag{4–2}$$

Equations 4–1indicates that the intermodal network emission consists of the carbon emissions of each single multimodal chain l from the hub node h ($h \in I$). Equations 4–2 indicates that in the time period p ($p \in P$) the sum of each chain transport flow $Q_{l,p}^{network}$ is equal to the shipment flow $Q_{h,p}^{network}$ of the hub node in the network. $X_{hd}^l = 1$ denotes that the transport route from the hub node h ($h \in I$) to the destination d is $l(l \in L)$,otherwise, $X_{hd}^l = 0$.

$$E_h^l = \sum_{(i,j)\in I} \sum_{v \in V_n} E_{ij}^v \cdot X_{ij}^v \cdot \left(1 - \delta_i^j\right) + \sum_{(i,j)\in I(n,g)\in N} \sum E_i^{n-g} \cdot Y_{n-g}^i \cdot \delta_i^j \tag{4–3}$$

$$\sum_{(i,j)\in I} X_{ij}^n = 1, \forall n \in N \tag{4-4}$$

$$v_{cap} \cdot X_{ij}^v \le Q_{ij}^n, \forall v \in V_n; \forall j,i \in I; \forall n \in N \tag{4–5}$$

Table 1
Definitions of sets.

Set	Definitions
L	the set of the intermodal transport chains l, $I \in L$.
I	the set of the intermodal transport nodes, $j,i,h,d \in I$.
A	the set of the energy $a \in A$.
N	the set of transport modes, $N = \{railway, road, inland\ water\ way\}$, and $n,g \in N$.
V_n	the set of transport vehicles v for mode $n, n \in N$
R	the set of transfer operation phases $r, r \in R$.
P	the set of time periods $p, p \in P$.
K	the set of equipment for loading and unloading operations $k, k \in K$.

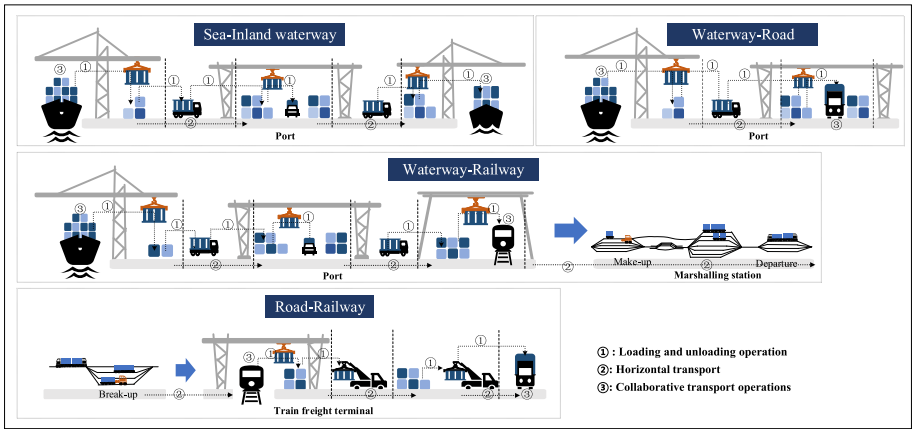


Fig. 4. Transshipment operations in intermodal terminals.

Where E_h^l is the carbon emissions of a single intermodal chain l from the hub node $h(h \in I)$ in Eqs. 4–3. E_{ij}^v is the carbon emissions from transport between transport node i to j by transport vehicle v, E_i^{n-g} is the carbon emission during the transfer of transport mode n to g. $X_{ij}^v = 1$ indicates transport vehicle v $(v \in V_n)$ transport between transport node i to j, otherwise, $X_{ij}^v = 0$. $Y_{n-g}^i = 1$ indicates transfer at node i from transport mode n to g, otherwise, $Y_{n-g}^i = 0$. When $i = j$, $\delta_i^j = 1$ indicates that the freight is in transfer at node i, and when $i \neq j$, $\delta_i^j = 0$ indicates that the freight is in transportation. Eqs. 4–4indicates that the connection between two nodes can only choose one transport mode n. Eqs. 4–5 indicates that the freight flow Q_{ij}^n cannot exceed the transport capacity v_{cap}.

4.2. Carbon emission model for multimodal transport processes

The carbon emissions mode of transport processes is as follows.

$$E_{ij}^v = ECF_{ij}^{v,a} \cdot F_a^v \tag{4–6}$$

Where $ECF_{ij}^{v,a}$ is the energy consumption of transport vehicle v from node i to j. F_a^v is the carbon emission factor for transport vehicle v in fuel a. The form of F_a^v depends on the unit of energy consumption: kg- CO_2/kg, kg-CO_2/MJ or kg- CO_2/kWh. If the energy unit is kg or kWh, the carbon emission factor requires conversion using the calorific value (IPCC, 2006).

$$ECF_{i,j}^{truck,a} = \left(EC^{emp} + \left(EC^{full} - EC^{emp}\right) \cdot lf_{truck}\right) \cdot D_{i,j} \cdot \tau_{em}^{truck} \cdot \tau_{rc}^{truck} \cdot Q_{ij}^{truck,a} \tag{4–7}$$

Eqs. 4–7 is the energy consumption for container road transport. Where EC^{emp}, EC^{full} $(MJ/tkm)$are respectively the energy consumption of truck for a given empty load and full load. $D_{i,j}$ is the highway mileage. lf_{truck} is the truck load factor which consists of the ratio of actual load to payload capacity (PC_{truck}). PC_{truck} denotes the maximum weight of freight that the transport vehicle can load. τ_{em} is the empty mileage coefficient, $\tau_{em}^{truck} = 1 + e_{em}^{truck}$, where e_{em}^{truck} denotes the empty mileage factor, the ratio of empty miles to loaded miles of trucks in $D_{i,j}$. τ_{rc}^{truck} is the road condition coefficient, $\tau_{rc}^{truck} = 1 + e_{rc}^{truck}$, where e_{rc}^{truck} denotes the road condition factor consisting of the ratio of ordinary road miles to highway miles and the ratio of resistance of different roads. $Q_{ij}^{truck,a}$ is the number of containers transported between node i and node j by truck using fuel a.

$$ECF_{i,j}^{rail,a} = ECF_{rail,a}^{ecp} \cdot NC_{train} \cdot D_{i,j} \cdot \tau_{em}^{rail} \cdot \tau_{rc}^{rail} \cdot Q_{ij}^{rail,a} \tag{4–8}$$

$$NC_{train} = \left(1 + \frac{w_{em}}{PC_{train} \cdot CU_{train}}\right) \tag{4–9}$$

Eqs. 4–8 is the energy consumption per unit based on historical statistics of locomotives in the railway sector. Where $ECF_{i,j}^{rail,a}$ is the energy consumption of per net tone kilometer of internal combustion or electric locomotives in the railway sector in $Wh/Ntkm$. $ECF_{rail,a}^{ecp}$ is the historical statistics of energy consumption per gross tone kilometer of internal combustion engines or electric locomotives in the railway sector in $Wh/Gtkm$. τ_{em}^{rail} and τ_{rc}^{rail} is the empty mileage coefficient, $\tau_{em}^{rail} = 1 + e_{em}^{rail}$. e_{em}^{rail} *is* empty mileage factor. τ_{rc}^{rail} is the railway condition coefficient. $\tau_{rc}^{rail} = 1 + e_{rc}^{rail}$. e_{rc}^{rail} denotes the road condition factor. NC_{train} is the net-gross relation factor. w_{em} is the empty weight of wagon (ECOTRANSIT, 2019). PC_{train} is the payload capacity of train. CU_{train} is the capacity utilization of train.

$$ECF_{i,j}^{ship,a} = SFC_{ME} \cdot P_{ME} \cdot (1 + \tau_{AE}) \cdot \tau_w \cdot \frac{D_{i,j}}{\bar{s}_{i,j}} \tag{4–10}$$

$$SFC_{ME} = SFC_{base} \cdot \left(0.45 \cdot load_{ship} - 0.71 \cdot load_{ship} + 1.280\right) \tag{4–11}$$

$$ECF_{ship}^{Q_{ij}^{ship,a}} = \frac{ECF_{i,j}^{ship,a}}{Q_{dwt}} \cdot Q_{ij}^{ship,a} \tag{4–12}$$

Eqs. 4–10 is the energy consumption for container inland waterway transport, P_{ME} and SFC_{ME} respectively represent the average main engine power and energy consumption per kW-hour during the ship's cruise. SFC_{ME} is the function of specific energy consumption and the engine load ratio for ship, as in Eqs. 4–11. In general, SFC_{base} indicates the specific fuel consumption of different engines and fuels at 80% MCR (maximum rated power). $load_{ship}$ denotes the load rate and takes the value 0–1. τ_{AE} is the energy consumption coefficient for auxiliary engines and boilers during the cruise. τ_w is resistance coefficient, $\tau_w = 1 + e_w$. e_w is the energy consumption resistance factor considering load, stream, and wind. $\bar{s}_{i,j}$ is the average speed of the ship. Eqs. 4–12is the energy consumption of ship transport for a given batch of containers $Q_{ij}^{ship,a}$. Q_{dwt} is the total number of containers loaded on the ship within the allowable deadweight tonnage.

4.3. Carbon emission model for transshipment processes in intermodal transport

$$E_i^{n-g} = EM_i \cdot Q_i + ECF_i \tag{4–13}$$

Eqs. 4–13 indicates that the carbon emission of the transshipment process consists of the unit energy consumption of transshipment operations (EM_i) and transport vehicles (ECF_i) at node i. Q_i represents the number of containers transiting at node i.

$$EM_i = \sum_{r \in R} \sum_{k \in K} E_r^k \cdot X_r^k \cdot F_a^k, \forall i \in I; \forall a \in A \tag{4–14}$$

$$E_r^k = e_r^k \cdot u_k^{-1} \cdot \tau_r^k \tag{4–15}$$

$$u_k = \frac{1}{W} \sum_{w \in W} \frac{Q_{i,w}}{T_{i,w}}, \forall i \in I \tag{4–16}$$

Eqs. 4–14 is the energy consumption per TEU for transfer operations at transfer nodes. $X_r^k = 1$ indicates that operation phase r uses equipment k, otherwise it is 0. F_a^k is the carbon emission factor for equipment k. E_r^k is the per unit energy consumption of equipment operation in Eqs 4–15. e_p^k is the operational energy consumption index. τ_r^k is the load factor for the empty-loading ratio. $\tau_r^k = (1 + \gamma)$, γ indicates the product of the empty-loading ratio of energy consumption and operational time. Equations 4–16 indicates that the operational efficiency (u_k) in TEU/h (Peng et al., 2020) is obtained from the mean value of the historical statistical period w batch operation time $(T_{i,w})$ and the flow of batch container operations $(Q_{i,w})$.

$$ECF_i = \sum_{v \in V_n} ECF_i^v \cdot t_i^v \cdot Q_v \cdot F_a^v, \forall i \in I; \forall a \in A \tag{4–17}$$

Eqs. 4–17is the energy consumption of the transport vehicle in port to assist the handling activities in port i, composed of the energy consumption per hour ECF_i^v, the operation time t_i^v, the number Q_v of the transport vehicle v, and the carbon emission factor F_a^v.

4.4. The flows distribution model for freight in intermodal networks

We combined previous studies of gravity models in freight flow allocation (Arbues and Banos, 2016; Levine et al., 2009; Shen and Aydin, 2014) with the problem of intermodal network flows in this paper. Considering the lack of historical freight flow data between OD pairs, this paper takes indicators that can represent for freight attractiveness of the network nodes. Then we incorporate indicators with a gravity model to obtain freight attractiveness values. Through the attractiveness value, we get the flow quota of each route.

We make the following assumptions for the model: (1) The model disregards traffic congestion and the OD pair flows of each transport mode are always distributed to the shortest route. (2) The total sending flow ($Q_{h,p}^{network}$) is equal to the receiving flow ($Q_{d,p}^{network}$), which is $Q_{h,p}^{network} = \sum_{d} Q_{d,p}^{network}$.

$$Q_{hj,p} = \varphi\left(R_{hj}, Q_{h,p}^{network}\right) \tag{4–18}$$

$$R_{hj} = \frac{T_h \cdot T_j}{f\left(d_{hj}^{n}\right)} \tag{4–19}$$

$$f\left(d_{hj}^{n}\right) = \sum_{n=1}^{N} \gamma_n \cdot t_{hj}^{n} \cdot c_{hj}^{n}, \forall h, j \in I \tag{4–20}$$

$$\varphi\left(R_{hj}, Q_{h,p}^{network}\right) = \frac{R_{hj}}{\sum_{l \in I_{ij}, j=1}^{j} R_{hj}} \cdot Q_{h,p}^{network}, \forall i \in I \tag{4–21}$$

$$Q_{hj,p}^{n} = \gamma_n \cdot Q_{hj,p} \tag{4–22}$$

Eqs. 4–18 indicates that $Q_{hj,p}$ is the flow distribution function composed of R_{hj}, $Q_{h,p}^{network}$. In Eqs. 4–19, R_{hj} represents the value of freight attractiveness from node city h ($h \in I$) to node city j ($j \in I$). T_h, T_j represent the indicators of freight flow of node h and node j. In Eqs. 4–20, $f(d_{hj}^{n})$ denotes the integrated distance function among nodes; γ_n is the proportion of transport mode n; t_{hj}^{n} is the shortest transport time of transport mode n from node city h ($h \in I$) to node city j ($j \in I$), and c_{hj}^{n} is the minimum transport cost per hour. Eqs. 4–21represents the flow quota ($Q_{hj,p}$) for each route in period p through the proportional value of the attractiveness distribution. Equations 4–22represents the freight flow ($Q_{hj,p}^{n}$) transported via transport mode n.

5. Case study and discussion - Yangtze River Delta transport network

This study takes the intermodal transport network of Shanghai port and the Yangtze River Delta hinterland as a case study. Based on the transportation network consisting of highways, railways and inland waterways in the YRD (Fig. 5(a–d)), and the freight data of 40 cities in the network from 2011 to 2021, we estimate the carbon emissions for the container transportation from the Shanghai port to the YRD hinterland in the past ten years. Shanghai Port is the largest container

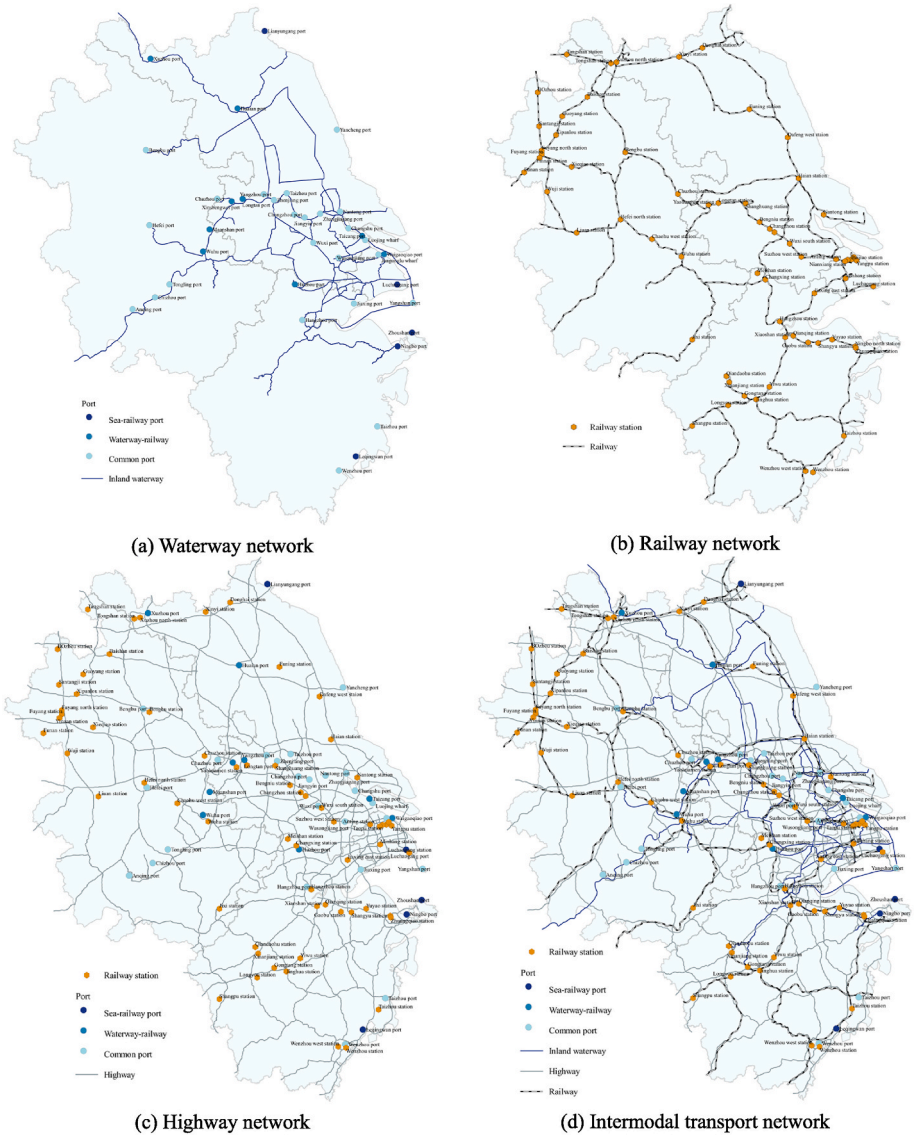


Fig. 5. Intermodal transport network in the YRD (by QGIS).

terminal around the world, which is located at the junction of the Yangtze River Economic Belt and the oversea maritime network. As an influential economic hinterland adjacent to the Shanghai port, the YRD Economic Zone is deeply connected to the Shanghai port both in financial and transportation.

5.1. Data specification

The data type of the case study includes network infrastructure and routes, freight flows, transport vehicles, transfer nodes, and carbon emission factors. These data come from survey data of companies and departments, government and industry reports, Baidu map open data, and previous literature. The detailed data descriptions and sources are shown in Table 2. Based on the collected data, we make a further specification as follows. (1) The network flow is a single-way flow of inbound containers from the port to the hinterland. (2) The annual container flows data of Shanghai port to the YRD are assigned to 40 cities according to the flow distribution model mentioned in section 3.4. (3) The last kilometer problem of transport to the port/station or the warehouse is solved by road. We assume that the final destination is the center of each city, and the driving distance is the distance from each port or station to the city center. (4) We make assumptions about the fuel choices available for different transport modes, road with diesel, electricity and LNG, rail with diesel and electricity, and ship with diesel and LNG. (5) The containers are all 40 feet TEU, with an average weight of 20t. The detailed data is shown in Table 3, Table 4, and Table 5.

Table 2
Data types and sources.

Data types	Data Description	Sources
Network infrastructure and routes	Rail freight yards; inland waterway ports; transportation distance of the shortest route for road, railway, and inland waterway; the share of transport structure.	Baidu map data through Python program; Schematic diagram of national railway freight business stations (2020); Railway odometers, National Highway Network Layout Plan (2009), Layout plan for the advanced waterway network in the YRD (2012); Yangtze River Waterway Mileage Table; Layout plan for national inland waterways and ports (2016–2020).
Freight flows	Freight flows; transport mode structure; transport time and costs.	Statistical Yearbook of 41cities in the YRD (2012–2021); 2020 Annual Report on Integrated Transport Development in the Yangtze River Delta, 2020 Bulletin of Transport Statistics.
Transport vehicles	Vehicle type and fuels; operation condition; load capacity; fuel choices proportion of transport vehicle.	2020 Bulletin of Transport Statistics; de Miranda Pinto et al. (2018); Ferreira et al. (2020); Hao et al. (2012; Heinold and Meisel (2018); Lin (2019); Zhang et al. (2017); Zhao et al. (2020); IMO (2020, 2014); Tao and Wu (2021); ECOTRANSIT (2019); Sun et al. (2013).
Transfer nodes	Type of operating equipment; energy choices of equipment; operating efficiency of equipment per hour.	Chen et al. (2018); Dai and Yang (2020); Fahdi et al. (2021); He et al. (2021); Hoang et al. (2022); Peng et al. (2020; Tichavska and Tovar (2015); L. Wang et al. (2020); Xiao et al. (2022).
Carbon emission factors	Carbon emission factor; net calorific value.	Altarazi et al., 2022; IPCC (2006); Ternel et al. (2021); China Energy Statistics Yearbook (2020).

5.2. Estimation results

The calculation results of the container flow distribution and the carbon emission estimation from 2011 to 2020 under the TTW and WTW modes are shown in Table 6 and Fig. 6. Meanwhile, as shown in Fig. 7-Fig. 8(a–b), we compared the container flows of different cities in 2020 and the corresponding carbon emissions under the WTW and TTW modes. Apparently, the carbon emissions in the case network are determined by the freight flows, transport distance and transshipment process. Taking 2020 as an example, the total carbon emissions of the network would be approximately 1.81 million tones- CO_2 (Mt- CO_2) under the WTW mode and 1.62 Mt- CO_2 under the TTW mode. As shown in Fig. 6, in the YRD intermodal network, Shanghai has the largest flow of containers, but its short transport distance puts the total carbon emissions in the middle level. In turn, Nanjing has the largest carbon emissions in the network due to the effect of freight flows, transport distances and transshipment.

It is also clear from Fig. 8(a)–(b) that the carbon emission differs between the different estimation modes, where the WTW mode has a higher carbon emission than the TTW mode. There are also differences in the distribution of carbon emissions based on the two models for different cities. The main reason for this gap is whether if electrical energy is involved in the emissions estimation process.

Based on the container flow allocation data in 2020, we analyzed the transport modes, the carbon emission, and the fuel choices distribution, as shown in Fig. 9. The distribution of container flows and carbon emissions maintain a high consistency. Sea to the road, inland waterway, and rail account for 59.3%, 38.9%, and 1.8% of the container flow, and the carbon emissions under WTW mode are 76.27%, 22.65%, and 1.1%, as well as keeping a similar ratio for TTW mode. The total carbon emissions intensity (the quotient of carbon emissions and the total container flow) was 0.199 t-CO_2/TEU in 2020. Carbon emissions from the transportation process account for 94.98%, and from the transit

Table 3
Attributes of the transport vehicle in this case.

Data stye	Values
Container size	40 feet
Actual weight of container $w_{container}$	20t/TEU
Payload capacity of truck PC_{truck}	26t/truck
Energy consumption of trucks in empty load ECF^{emp}	7.8 MJ/km
Energy consumption of trucks in full load ECF^{full}	10.6 MJ/km
Load factor of truck lf_{truck}	76.9%
Energy consumption of railway of diesel locomotives $ECF^{ecp}_{railway,\text{diesel}}$	79.7 Wh/Ntkm
Energy consumption of railway of electric locomotives $ECF^{ecp}_{railway,\text{electric}}$	21.1 Wh/Ntkm
Net-gross relation factor of train NC_{train}	0.52
Deadweight tonnage of ship Q_{dwt}	8311t
Engine load ratio for the type of ship $load_{ship}$	80%
Specific fuel consumption of ship SFC_{base}	175 g/kWh
Average main engine power during the ship's cruise P_{ME}	1384 kW
Energy consumption per kW-hour during the ship's cruise SFC_{ME}	188 g/kWh
Energy consumption factor for auxiliary engines and boilers in the cruise τ_{AE}	5%
Average speed of the ship $\bar{s}_{i,i+1}$	10.5kn/h
The empty mileage factor e^{truck}_{em}	10%
The railway empty mileage coefficient e^{rail}_{em}	20%
The resistance factor e_w	25%
The share of transport structure for road, railway, and inland waterway	59.3%:1.8%:38.9%
The fuel choices proportion of rail for diesel and electricity	25.1%:76.4%
The fuel choices proportion of road for diesel, electricity, and LNG	82.5%:15.6%:2.9%
The fuel choices proportion of road for inland waterway diesel and LNG	96.7%:3.3%

Table 4
Equipment operation data for nodes.

Equipment for nodal operations	Hourly Energy consumption e_r^k (kWh/h or kg/h)	Average equipment operating efficiency per hour u_k (TEU/h)
Quay crane	30.301	25 (25–30)
Yard crane	17.922	15 (15–25)
Yard trucks	11.803	10 (10–15)
Reach stacker	18.578	15 (15–20)
Rail-mounted gantry Crane	30.301	25 (25–35)

Table 5
Carbon emission factors and net calorific values for different fuel types.

Fuel types	Net calorific value	Carbon emission factors	
		TTW	WTW
Diesel	43 MJ/kg	74.1 g-CO_2/MJ	76 g- CO_2/MJ
Biodiesel	19.3 MJ/kg	70.8 g-CO_2/MJ	72 g- CO_2/MJ
Liquid Natural Gas	44.2 MJ/kg	64.2 g-CO_2/MJ	69 g- CO_2/MJ
Electricity	0.0036 MJ/kWh	0	583.9 g- CO_2/kWh

process account for 5.02%. Diesel, electricity, and LNG account for 95.47%, 3.35%, and 1.18% of the fuel choices.

The case results show that although the transport and economic advantages of the YRD provide more opportunities for multimodal transport development, road transport and its carbon emissions still account for a significant share of the overall network, while diesel is still the main choice of energy at present. The lower carbon intensity of sea-rail intermodal transport is not prominent in the YRD network for the following reasons. For one thing, container flows share of sea-rail intermodal transport only accounts for 1.8% of the total flows. For another, the high percentage of transfer carbon emissions makes it challenging to take full advantage of the railway. As shown in Fig. 9, the carbon emissions of the sea-rail transfer process account for 34.84% of the total carbon emissions, while the sea-rail and sea-water only account for 0.66% and 1.33%.

5.3. Identification of driver factors on the carbon emission

The complexity of the carbon emission estimation model leads to a large number of parameters required for the model, and uncertainty in the parameters can affect the final results of the multimodal network estimation (Tao and Wu, 2021). The main uncertainties in this study are the container flow ratio and container weight of the freight flows forecast (A), vehicle activity efficiency (B), energy efficiency (C), transshipment efficiency (D), and port operation (E). To discuss the impact of uncertainty, we analyze the impact of different parameter changes on the overall carbon emissions, using container flows data in 2020 as the baseline value, and identify the critical driver factors affecting the network carbon emissions.

As shown in Table 7 and Fig. 10, the sensitivity degrees of WTW and TTW carbon emissions in the base year remain highly consistent for the majority, but some differences deserve special attention. The WTW mode shows a significant reduction in energy efficiency improvements of truck (B1) with a sensitivity factor of 80.78%, leading the group. Energy efficiency represents the improvement of vehicle and vessel energy consumption and carbon emission technologies, with a 1% improvement resulting in 17,200 tons of carbon emission reduction. The next is the increase in container weight (A3), with a sensitivity factor of 60.11%. Container weight represents the optimal load distribution of containers. Finally, the transfer of road container flows to rail and water transport has a sensitivity factor of 43.02% and 41.82%, respectively.

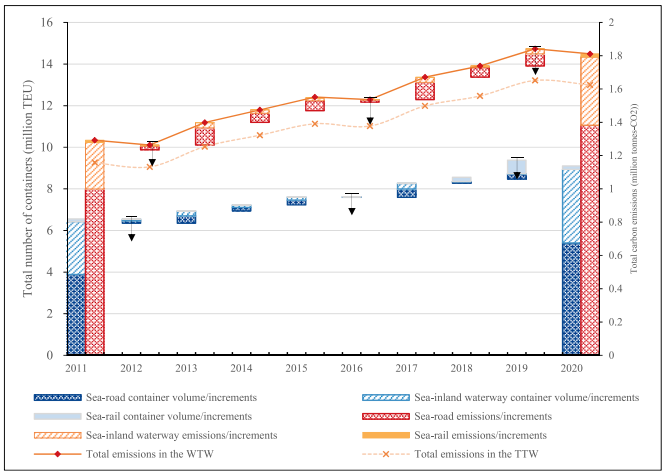


Fig. 6. Case network container flows and total carbon emissions from 2011 to 2020.

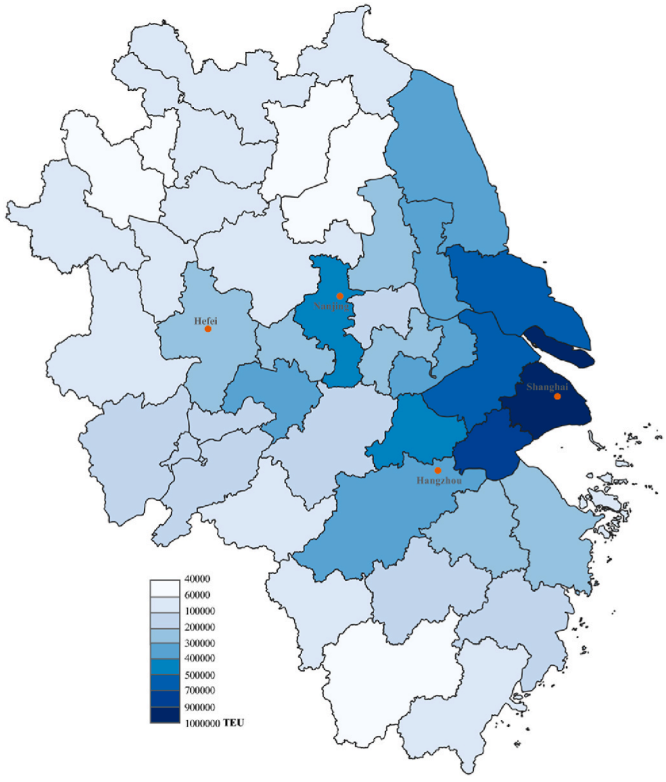
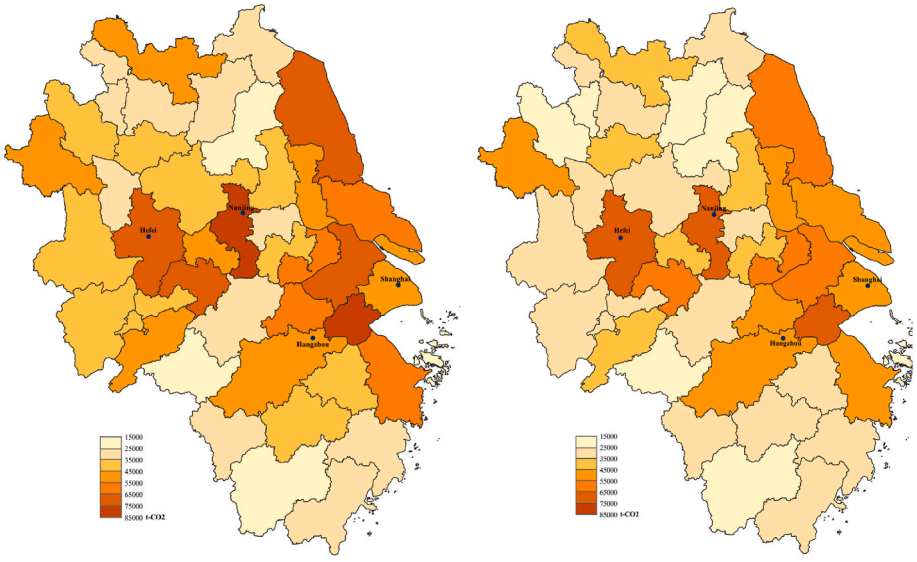


Fig. 7. Container flows distribution for the 2020 case network.

Table 6
Container flows and carbon emission in case intermodal network.

Data	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Container flow (Million TEU)	6.55	6.35	6.94	7.23	7.62	7.59	8.27	8.47	9.12	9.10
Total carbon emissions in the WTW mode of network (Mt- CO_2)	1.29	1.26	1.40	1.47	1.55	1.54	1.67	1.74	1.84	1.81
Total carbon emissions in the TTW mode of network (Mt- CO_2)	1.16	1.13	1.25	1.32	1.39	1.38	1.50	1.56	1.65	1.62



(a) Carbon emissions of the WTW mode (b) Carbon emissions of the TTW mode

Fig. 8. Carbon emissions distribution from the case network in 2020.

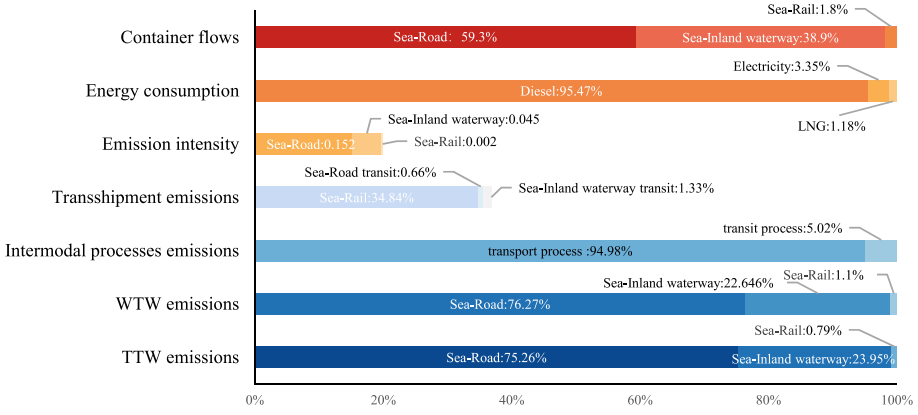


Fig. 9. Structure of case data and results.

The transfer of 10% of container flows from road transport to rail and water transport at the same time can bring 153,600 tons of carbon emission reduction, accounting for 8.4% of the total emissions in 2020. As seen in Fig. 8, rail and water transport have the share of container flows and low carbon intensity in the current network, which indicates that there is still considerable potential to adjust the existing transport structure for emissions reduction.

What is different in the TTW mode is that the shift of truck energy to electricity C2 shows a higher sensitivity, just after B2, with a sensitivity factor of 71.56%. In contrast, in the WTW mode, where the production process of electric power participates in the carbon emissions estimation, the sensitivity factor of C2 is 25.84%, which is far lower than shown in the TTW mode. Thus, the impact of electricity is far more significant in the WTW mode than in the TTW mode (Gustafsson et al., 2021). And we should pay more attention to the fact that different modes of carbon emissions estimation will influence preferences for making decisions about emissions reduction.

The transfer process is an essential medium for connecting intermodal transport. Reducing emissions from the transfer process can promote the transition from high carbon-emitting transport to low ones. In the enlarged graph of Fig. 10, the increase of transit efficiency is more sensitive in both estimation modes. The sensitivity factor for the WTW mode in 2020 is 4.6%, with a 10% increase in efficiency resulting in a 0.84 Mt-CO_2 decrease in carbon emissions, accounting for 0.5% of total carbon emissions and 9.3% of transit emissions in 2020. It is followed by an improvement in the proportion of ship-truck direct transfers, with a sensitivity factor of 0.1%. Therefore, improving the overall efficiency of the transfer nodes and optimizing the truck-ship transfer operation have an important effect on the carbon emission reduction in the transfer process.

Based on the sensitivity analysis above, even the best way is to improve the truck energy efficiency to reduce the carbon emission, it's not easy for carriers to choose that because of the decreased engine efficiency and higher maintenance costs. In contrast, shifting road transport and container loading are better choices for the transportation department. So, we rank the critical driver factors according to the operability of the transport sector and intermodal companies on emissions reduction as follows: shifting of road transport to rail and road, increasing container weight, applying clean energy, improving truck energy efficiency, improving transfer efficiency.

5.4. Scenario analysis based on carbon reduction policies

To analyze the impact of emission reduction policies on CO2 emissions, we group the driver factors mentioned above into four categories as follows: transport restructuring (W1), clean energy application (W2),

Table 7
Sensitivity analysis parameters in cases.

Transport process parameters		WTW		TTW	
		Ratio of decrease	Sensitivity factor	Ratio of decrease	Sensitivity factor
A	A1: 10% shift in flows from road to rail	4.302%	43.024%	4.916%	49.155%
	A2: 10% shift in flows from road to water	4.182%	41.821%	3.874%	38.744%
	A3:10% increase in container weight	6.011%	60.113%	6.142%	61.420%
B	B1:1% increase in energy efficiency for trucks	0.808%	80.780%	0.794%	79.415%
	B2:1% increase in energy efficiency by rail	0.004%	0.412%	0.002%	0.167%
	B3:1% increase in energy efficiency of ships	0.139%	13.903%	0.151%	15.087%
C	C1:10% shift from diesel to LNG for trucks	0.586%	5.861%	0.988%	9.877%
	C2: 10% shift from diesel to electricity for trucks	2.584%	25.837%	7.156%	71.564%
	C3: 10% shift from diesel to electricity for railways	0.007%	0.067%	0.050%	0.498%
	C4: 10% shift from diesel to LNG for waterways	0.086%	0.863%	0.160%	1.602%
Transfer process parameters		WTW Ratio of decrease	Sensitivity factor	TTW Ratio of decrease	Sensitivity factor
D	D1:10% increase in transfer efficiency	0.460%	4.601%	0.170%	1.696%
E	E1:10% increase in rail access ratio	0.009%	0.085%	0.009%	0.093%
	E2:10% increase in direct ship-train transfer	0.028%	0.282%	0.031%	0.306%
	E3:10% increase in direct train-truck transfer	0.001%	0.015%	0.002%	0.016%
	E4:10% increase in direct ship-truck transfer	0.100%	0.995%	0.108%	1.082%

*Sensitivity factor = Ratio of decrease/Rate of change.

energy efficiency for transport vehicles (W3), and transfer technology upgrading (W4). And we use it as critical indicators for different policy scenarios to analyze the carbon emission trends under different scenarios. The specific scenario descriptions and assumptions are shown in Table 8, where we assume parameters based on the national policy instrument on emission reduction planning (General Office of the State Council of the PRC, 2021; The Communist Party of China Central Committee and the State Council, 2021; The State Council of the PRC, 2021b, 2021a).

The results and trends of carbon emissions in the case network are significantly different for the various scenarios, as shown in Table 9 and Fig. 11. In Scenario 1 of the WTW mode, the parameters variation remains at a low level. And the overall carbon emission maintains an increasing trend to 2060, at approximately 2.43 Mt-CO_2. In Scenario 2, the parameters vary according to the national policy instrument on emission reduction planning. In S2, the carbon emissions of the case

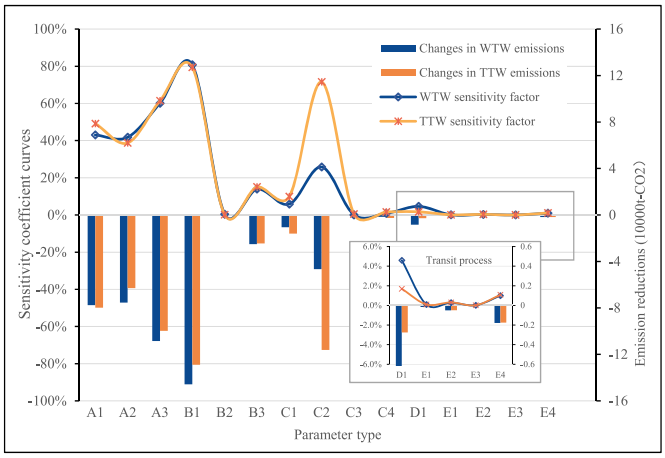


Fig. 10. Sensitivity analysis of key parameters of the case.

Table 8
Explanation of scenario situations.

	Scenario 1(S1)	Scenario 2(S2)-baseline	Scenario 3(S3)
Scenario status	Trade blocked, brake in economic development, the completion of planning targets will be negatively impacted. 5% reduction in freight flows from the baseline value.	The economic and trade growth at the current rate; the national planning target will be completed on schedule. The freight flows are regression forecasts based on data from 2011 to 2020.	A more favorable economic environment, rapid trade growth, and emphasis on environmental protection and climate change.5% increase in freight flows from the baseline value.
W1	A 10% annual increase in sea-rail flows to 2035, slow growth thereafter, and a 5% increase in sea-water flows to 2025 and then maintaining the ratio.	A 15% annual increase in sea-rail flows to 2035 and slow growth after that, and a 10% increase in sea-water flows to 2025 and then maintaining the ratio.	A 20% annual increase in sea-rail flows to 2035, slow growth thereafter, and a 15% increase in sea-water flows to 2025 and then maintaining the ratio.
W2	Electricity carbon intensity declining at 2% per 5 years to meet EU levels by 2060; non-fossil fuel permeability 5% per 5 years to 50% by 2060	Electricity carbon intensity declining at 5% every 5 years to 70% by 2060 and non-fossil fuel penetration at 10% every 5 years to 70% by 2060.	Electricity carbon intensity declining at 8% every 5 years to 80% by 2060 and non-fossil fuel penetration at 10% every 5 years to 90% by 2060.
W3	0.2% improvement in energy efficiency every 5 years, 5% improvement to 2060.	0.5% improvement in energy efficiency every 5 years, 20% improvement to 2060.	0.8% improvement in energy efficiency every 5 years, 30% improvement to 2060.
W4	5% increase in the proportion of rail access and new energy in nodes every 5 years to 40% in 2060; 3% increase in transshipment efficiency every 5 years to 40% in 2060.	10% increase in the proportion of rail access and new energy in nodes every 5 years to 60% in 2060; 5% increase in transshipment efficiency every 5 years to 60% in 2060.	15% increase in the proportion of rail access and new energy in nodes every 5 years to 90% in 2060; 10% increase in transshipment efficiency every 5 years to 90% in 2060.

network will peak during the period 2030–2035 and then decrease gradually to 1.4 Mt-CO_2 in 2060. Scenario 3 with an aggressive level of parameter variation will peak before 2030 and then decreases rapidly to only about 0.88 Mt-CO_2 in 2060. Scenario 2 confirms the validity of the

Table 9
Carbon emission results for different scenarios.

Modes	Scenario	2025	2030	2035	2060
WTW	Carbon emissions in S1 (kg)	1906480	2097949	2249082	2427644
	Change rate of S1	5.28%	10.04%	7.20%	15.72%
	Carbon emissions in S2 (kg)	1941238	2044376	1877441	1403967
	Change rate of S2	7.20%	5.31%	−8.17%	−31.33%
	Carbon emissions in S3 (kg)	1943178	1913262	1682711	883369
	Change rate of S3	7.31%	−1.54%	−12.05%	−53.83%
TTW	Carbon emissions in S1 (kg)	1673543	1790302	1866065	1928548
	Change rate of S1	3.02%	6.98%	4.23%	7.72%
	Carbon emissions in S2 (kg)	1665976	1667105	1422509	1194976
	Change rate of S2	2.56%	0.07%	−14.67%	−28.32%
	Carbon emissions in S3 (kg)	1615690	1461975	1146548	634289
	Change rate of S3	−0.54%	−9.51%	−21.58%	−56.61%

current national policy planning for achieving carbon peaking in the case network. The TTW mode follows the same trend as the WTW mode but peaks earlier, peaking around 2030 in S2 and around 2025 in S3. This phenomenon is because electricity, as zero-carbon emission in the TTW mode, shares 5% (in Fig. 9) and even more (after parameters variation) of carbon emissions, and this part of carbon emissions are ignored.

The carbon emission variation by parameter changes in the baseline S2 under the WTW mode is presented in Fig. 11. The emission reductions caused by W1 and W2 are remarkable compared to without any measure, bringing 19.27% and 13.36% reduction respectively under Scenario 2 in 2035, and bringing 29.87% and 38.39% in 2060.

We take the data from Scenario 2 under the WTW mode in 2035 as an example and present a portrait of data distribution after decoupling, as shown in Fig. 12. Of these, the intermodal transport structure of sea-rail, sea-water, and sea-rail account for 51.23%, 32.43%, and 16.34%, respectively. The fuel choices of diesel and biodiesel, electricity, and LNG account for 63.08%, 23.74%, and 8.17%. The carbon emission intensity is 0.134t-CO_2/TEU for the whole network, 0.0688t-CO_2/TEU for sea-road intermodal transport, 0.022t-CO_2/TEU for sea-rail, and 0.044t-CO_2/TEU for sea-water. Compared with 2020 (in Fig. 9), the whole network emission intensity decreased by 32.5%. Scenario 2 shows a minor reduction in energy efficiency parameters for transport vehicles (W3) and transfer technology upgrading (W4).

6. Conclusion and management insights

This study takes the port as the intermodal hub and establishes the intermodal container network based on the port container flow. Through the process analysis, we discuss the carbon emission model of multimodal transport based on the transport and transfer processes. Then we develop a hinterland-based carbon emission model for multimodal transport network and a network flow distribution model to evaluate the carbon emission of the intermodal transport network. Finally, as an example, we discuss the carbon emissions of the intermodal container network of Shanghai port to the YRD hinterland. This study provides a novel approach and a broader perspective for studying carbon emissions in intermodal transport. The main findings are as follows.

- Road transport and its carbon emissions still account for a significant share of the overall network, accounting for 59.3% of the container flow and 76.27% of the total carbon emissions in the case network,

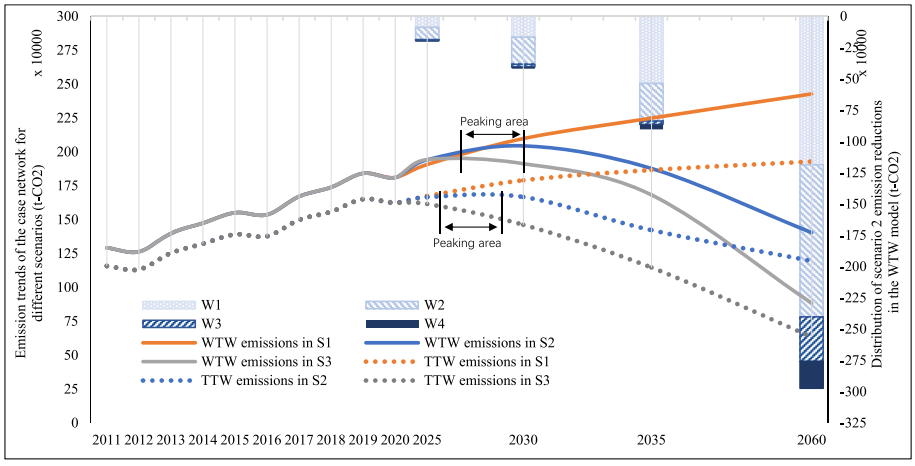


Fig. 11. Trends in carbon emissions for different scenarios of the case network.

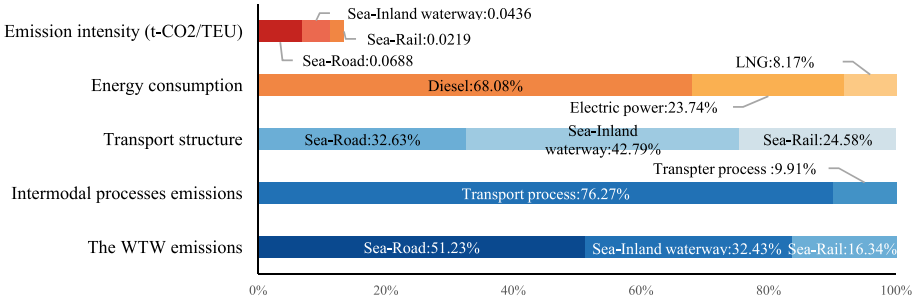


Fig. 12. Distribution of 2035 case data and results in Scenario 2 with the WTW mode.

respectively. Diesel consumption (92.07%) is still the main contributor to energy consumption.

- Sea-rail intermodal transport, which has the lowest carbon intensity in the network, does not feature prominently in the YRD network due to the poor transfer process and a small share of freight transport.
- The transition of transport structure and fuel choices significantly affects the overall emission reduction, followed by container weight.
- Based on the current policy scheme scenario, the case network will peak in 2030–2035 at 2 million tons of CO_2 under the WTW mode. The TTW mode will peak earlier than the WTW mode in 2030. The overall network carbon emission intensity is 0.1343t-CO_2/TEU, a 32.5% decrease compared to 2020.
- The efficiency of equipment operation and the organization of vehicle-vessel transit impact the transfer carbon emission significantly. Their contribution to emissions reduction is minor compared to that of the entire network.

Two managerial insights are included as follows:

- For the government, the transition of the transport to rail and water together is more operational for carbon emission reduction than energy technological upgrading of the road. Optimizing the transport structure makes it more accessible to achieve carbon peaks sooner. The government should provide more incentives for transport restructuring. There should also be attentive to the optimal proportion of different transport modes (Bouchery and Fransoo, 2015). And the recommended target ratios can be standardized by the case network date in 2035, with a balance of 5:3:2 for road, water, and rail. For the energy department, with the current high carbon intensity of the national electricity grid (0.5839 kg-CO_2/kWh), the analysis of network carbon emissions under the WTW mode indicates that electricity consumption of the transport sector is necessary to share the responsibility for carbon emissions from energy production. This view provides an insight into the development of carbon tax policies in the future. Taking into account the extensive share of diesel fuel in transport, we introduce biofuels with lower carbon emissions as a transitional way to the shift from traditional diesel to clean fuel (Yazan S M Altarazi et al., 2022; Ternel et al., 2021).
- For multimodal transport enterprises, they should improve container capacity utilization by optimizing transport organization and enhancing the loading ratio. Enterprises should also cooperate with government policies to reduce emissions, such as upgrading operational equipment, promoting transport vehicles with clean energy.

Further work in this research will focus on the following: in this study, the data for several parameters in the carbon emission model are referenced from previous literature. It is necessary to estimate them from actual operational data. Furthermore, carbon peak should be quantified by estimation models rather than only by the trend analysis. We will explore these works in future studies to improve the carbon emission theory of multimodal transport networks.

CRediT authorship contribution statement

Xiaoyan Guo: Conceptualization, Formal analysis, Methodology, Writing – original draft. **Junliang He:** Formal analysis, Writing – original draft, Resources, Writing – review & editing, Project administration, Funding acquisition. **Mo Lan:** Data curation, Investigation, Writing – review & editing. **Hang Yu:** Writing – original draft, Writing – review & editing. **Wei Yan:** Conceptualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgement

This work is sponsored by the National Natural Science Foundation of China [grant numbers 72072112, 72001135, 72002125]; Natural Science Foundation of Shanghai [grant number 22ZR1426600]; Shanghai Sailing Program [grant number 20YF1416600]; Shanghai Science and Technology Committee project [grant numbers 21010501800, 22010501900]; China Association for Science and Technology Young Talent Program [grant number 2022ZZ041560].

References

Altarazi, Yazan S.M., Abu Talib, A.R., Yu, J., Gires, E., Abdul Ghafir, M.F., Lucas, J., Yusaf, T., 2022. Effects of biofuel on engines performance and emission characteristics: a review. Energy 238. https://doi.org/10.1016/j.energy.2021.121910.

Altarazi, Yazan S.M., Abu Talib, A.R., Yu, J., Gires, E., Abdul Ghafir, M.F., Lucas, J., Yusaf, T., 2022. Effects of biofuel on engines performance and emission characteristics: a review. Energy 238, 121910. https://doi.org/10.1016/j.energy.2021.121910.

Arbues, P., Banos, J.F., 2016. A dynamic approach to road freight flows modeling in Spain. Transportation 43, 549–564. https://doi.org/10.1007/s11116-015-9590-5.

Bouchery, Y., Fransoo, J., 2015. Cost, carbon emissions and modal shift in intermodal network design decisions. Int. J. Prod. Econ. 164, 388–399. https://doi.org/10.1016/j.ijpe.2014.11.017.

Cariou, P., Parola, F., Notteboom, T., 2019. Towards low carbon global supply chains: a multi-trade analysis of CO2 emission reductions in container shipping. Int. J. Prod. Econ. 208, 17–28. https://doi.org/10.1016/j.ijpe.2018.11.016.

Chen, H., Lam, J.S.L., Liu, N., 2018. Strategic investment in enhancing port-hinterland container transportation network resilience: a network game theory approach. Transp. Res. Part B Methodol. 111, 83–112. https://doi.org/10.1016/j.trb.2018.03.004.

Chocholac, J., Hruska, R., Machalik, S., Sommerauerova, D., Krupka, J., 2021. Customized Approach to Greenhouse Gas Emissions Calculations in Railway Freight Transport, vol. 11. Applied Sciences, Switzerland. https://doi.org/10.3390/app11199077.

Craig, A.J., Blanco, E.E., Sheffi, Y., 2013. Estimating the CO2 intensity of intermodal freight transportation. Transp Res D Transp Environ 22, 49–53. https://doi.org/10.1016/j.trd.2013.02.016.

Cui, L., Mu, Y., Shen, Z., Wang, W., 2022. Energy transition, trade and green productivity in advanced economies. J. Clean. Prod. 361, 132288 https://doi.org/10.1016/j.jclepro.2022.132288.

Dai, Q., Yang, J., 2020. A distributionally robust chance-constrained approach for modeling demand uncertainty in green port-hinterland transportation network optimization. Symmetry (Basel) 12. https://doi.org/10.3390/sym12091492.

de Miranda Pinto, J.T., Mistage, O., Bilotta, P., Helmers, E., 2018. Road-rail intermodal freight transport as a strategy for climate change mitigation. Environ Dev 25, 100–110. https://doi.org/10.1016/j.envdev.2017.07.005.

Dreier, D., Silveira, S., Khatiwada, D., Fonseca, K.V.O., Nieweglowski, R., Schepanski, R., 2018. Well-to-Wheel analysis of fossil energy use and greenhouse gas emissions for conventional, hybrid-electric and plug-in hybrid-electric city buses in the BRT system in Curitiba, Brazil. Transp Res D Transp Environ 58, 122–138. https://doi.org/10.1016/j.trd.2017.10.015.

ECOTRANSIT, 2019. Ecological Transport Information Tool for Worldwide Transports.

Fahdi, S., Elkhechafi, M., Hachimi, H., 2021. Machine learning for cleaner production in port of Casablanca. J. Clean. Prod. 294, 126269 https://doi.org/10.1016/j.jclepro.2021.126269.

Fan, Y. van, Perry, S., Klemeš, J.J., Lee, C.T., 2018. A review on air emissions assessment: Transportation. J. Clean. Prod. 194, 673–684. https://doi.org/10.1016/j.jclepro.2018.05.151.

Ferreira, H., Rodrigues, C.M., Pinho, C., 2020. Impact of road geometry on vehicle energy consumption and CO2 emissions: an energy-efficiency rating methodology. Energies 13. https://doi.org/10.3390/en13010119.

General Office of the State Council of the PRC, 2021. Work Plan for Promoting the Development of Multimodal Transport to Optimize and Adjust the Transport Structure (2021-2025). China.

Ghane-Ezabadi, M., Vergara, H.A., 2016. Decomposition approach for integrated intermodal logistics network design. Transp Res E Logist Transp Rev 89, 53–69. https://doi.org/10.1016/j.tre.2016.02.009.

Gustafsson, M., Svensson, N., Eklund, M., Dahl Öberg, J., Vehabovic, A., 2021. Well-to-wheel greenhouse gas emissions of heavy-duty transports: influence of electricity carbon intensity. Transp Res D Transp Environ 93, 102757. https://doi.org/10.1016/j.trd.2021.102757.

Hao, H., Wang, H., Ouyang, M., 2012. Fuel consumption and life cycle GHG emissions by China's on-road trucks: future trends through 2050 and evaluation of mitigation measures. Energy Pol. 43, 244–251. https://doi.org/10.1016/j.enpol.2011.12.061.

He, J., Wang, Y., Tan, C., Yu, H., 2021. Modeling berth allocation and quay crane assignment considering QC driver cost and operating efficiency. Adv. Eng. Inf. 47, 101252 https://doi.org/10.1016/j.aei.2021.101252.
Heinold, A., Meisel, F., 2018. Emission rates of intermodal rail/road and road-only transportation in Europe: a comprehensive simulation study. Transp Res D Transp Environ 65, 421–437. https://doi.org/10.1016/j.trd.2018.09.003.
Hoang, A.T., Foley, A.M., Nižetić, S., Huang, Z., Ong, H.C., Ölçer, A.I., Pham, V.V., Nguyen, X.P., 2022. Energy-related approach for reduction of CO2 emissions: a critical strategy on the port-to-ship pathway. J. Clean. Prod. https://doi.org/10.1016/j.jclepro.2022.131772.
IEA, 2021. Greenhouse Gas Emissions from Energy: Overview. Paris.
IMO, 2020. Fourth IMO GHG Study 2020 Full Report.
IMO, 2014. Third IMO GHG Study 2014 Executive Summary and Final Report.
IPCC, 2006. 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
Jiang, J., Zhang, D., Meng, Q., Liu, Y., 2020. Regional multimodal logistics network design considering demand uncertainty and CO2 emission reduction target: a system-optimization approach. J. Clean. Prod. 248, 119304 https://doi.org/10.1016/j.jclepro.2019.119304.
Kaack, L.H., Vaishnav, P., Morgan, M.G., Azevedo, I.L., Rai, S., 2018. Decarbonizing intraregional freight systems with a focus on modal shift. Environ. Res. Lett. https://doi.org/10.1088/1748-9326/aad56c.
Lättilä, L., Henttu, V., Hilmola, O.-P., 2013. Hinterland operations of sea ports do matter: dry port usage effects on transportation costs and CO2 emissions. Transp Res E Logist Transp Rev 55, 23–42. https://doi.org/10.1016/j.tre.2013.03.007.
Levine, B., Nozick, L., Jones, D., 2009. Estimating an origin-destination table for US imports of waterborne containerized freight. Transp Res E Logist Transp Rev 45, 611–626. https://doi.org/10.1016/j.tre.2008.11.001.
Li, L., Zhang, X., 2020. Integrated optimization of railway freight operation planning and pricing based on carbon emission reduction policies. J. Clean. Prod. 263, 121316 https://doi.org/10.1016/j.jclepro.2020.121316.
Lin, N., 2019. CO2 emissions mitigation potential of buyer consolidation and rail-based intermodal transport in the China-Europe container supply chains. J. Clean. Prod. 240, 118121 https://doi.org/10.1016/j.jclepro.2019.118121.
Macharis, C., Bontekoning, Y.M., 2004. Opportunities for OR in intermodal freight transport research: a review. Eur. J. Oper. Res. 153, 400–416. https://doi.org/10.1016/S0377-2217(03)00161-9.
Miklautsch, P., Woschank, M., 2022. A framework of measures to mitigate greenhouse gas emissions in freight transport: systematic literature review from a Manufacturer's perspective. J. Clean. Prod. 366, 132883 https://doi.org/10.1016/j.jclepro.2022.132883.
Peng, Y., Liu, H., Li, X., Huang, J., Wang, W., 2020. Machine learning method for energy consumption prediction of ships in port considering green ports. J. Clean. Prod. 264, 121564 https://doi.org/10.1016/j.jclepro.2020.121564.
Pizzol, M., 2019. Deterministic and stochastic carbon footprint of intermodal ferry and truck freight transport across Scandinavian routes. J. Clean. Prod. 224, 626–636. https://doi.org/10.1016/j.jclepro.2019.03.270.
Ramalho, M.M., Santos, T.A., 2021. Numerical modeling of air pollutants and greenhouse gases emissions in intermodal transport chains. J. Mar. Sci. Eng. 9 https://doi.org/10.3390/jmse9060679.
Shen, G., Aydin, S.G., 2014. Origin-destination missing data estimation for freight transportation planning: a gravity model-based regression approach. Transport. Plann. Technol. 37, 505–524. https://doi.org/10.1080/03081060.2014.927665.
Sun, X., Yan, X., Wu, B., Song, X., 2013. Analysis of the operational energy efficiency for inland river ships. Transp Res D Transp Environ 22, 34–39. https://doi.org/10.1016/j.trd.2013.03.002.
Sun, Y., Yu, N., Huang, B.L., 2022. Green road-rail intermodal routing problem with improved pickup and delivery services integrating truck departure time planning under uncertainty: an interactive fuzzy programming approach. COMPLEX & INTELLIGENT SYSTEMS 8, 1459–1486. https://doi.org/10.1007/s40747-021-00598-1.
Talley, W.K., Ng, M., 2018. Hinterland transport chains: a behavioral examination approach. Transp Res E Logist Transp Rev 113, 94–98. https://doi.org/10.1016/j.tre.2018.03.001.
Tao, X., Wu, Q., 2021. Energy consumption and CO2 emissions in hinterland container transport. J. Clean. Prod. 279, 123394 https://doi.org/10.1016/j.jclepro.2020.123394.
Ternel, C., Bouter, A., Melgar, J., 2021. Life cycle assessment of mid-range passenger cars powered by liquid and gaseous biofuels: comparison with greenhouse gas emissions of electric vehicles and forecast to 2030. Transp Res D Transp Environ 97. https://doi.org/10.1016/j.trd.2021.102897.
The Communist Party of China Central Committee and the State Council, 2021. Working Guidance for Carbon Dioxide Peaking and Carbon Neutrality in Full and Faithful Implementation of the New Development Philosophy. China.
The State Council of the PRC, 2021a. The Action Plan to Peak Carbon Dioxide Emissions before 2030.
The State Council of the PRC, 2021b. A Modern Comprehensive Transportation System over the 14th Five-Year Plan (2021-2025.
Tichavska, M., Tovar, B., 2015. Environmental cost and eco-efficiency from vessel emissions in Las Palmas Port. Transp Res E Logist Transp Rev 83, 126–140. https://doi.org/10.1016/j.tre.2015.09.002.
Wang, L., Peng, C., Shi, W., Zhu, M., 2020. Carbon dioxide emissions from port container distribution: spatial characteristics and driving factors. Transp Res D Transp Environ 82. https://doi.org/10.1016/j.trd.2020.102318.
Wang, W., Xu, X., Jiang, Y., Xu, Y., Cao, Z., Liu, S., 2020. Integrated scheduling of intermodal transportation with seaborne arrival uncertainty and carbon emission. Transp Res D Transp Environ 88, 102571. https://doi.org/10.1016/j.trd.2020.102571.
Wiercx, M., van Kalmthout, M., Wiegmans, B., 2019. Inland waterway terminal yard configuration contributing to sustainability: modeling yard operations. Res. Transport. Econ. 73, 4–16. https://doi.org/10.1016/j.retrec.2019.02.001.
Xiao, G., Wang, T., Chen, X., Zhou, L., 2022. Evaluation of ship pollutant emissions in the ports of Los Angeles and Long beach. J. Mar. Sci. Eng. 10 https://doi.org/10.3390/jmse10091206.
Zhang, M., Wiegmans, B., Tavasszy, L., 2013. Optimization of multimodal networks including environmental costs: a model and findings for transport policy. Comput. Ind. 64, 136–145. https://doi.org/10.1016/j.compind.2012.11.008.
Zhang, Y., Gu, J., Wang, W., Peng, Y., Wu, X., Feng, X., 2017. Inland port vessel emissions inventory based on ship traffic emission assessment model-Automatic identification system. Adv. Mech. Eng. 9 https://doi.org/10.1177/1687814017712878.
Zhao, J., Zhang, Y., Patton, A.P., Ma, W., Kan, H., Wu, L., Fung, F., Wang, S., Ding, D., Walker, K., 2020. Projection of ship emissions and their impact on air quality in 2030 in Yangtze River delta, China. Environ. Pollut. 263 https://doi.org/10.1016/j.envpol.2020.114643.
Zhou, T., Roorda, M.J., MacLean, H.L., Luk, J., 2017. Life cycle GHG emissions and lifetime costs of medium-duty diesel and battery electric trucks in Toronto, Canada. Transp Res D Transp Environ 55, 91–98. https://doi.org/10.1016/j.trd.2017.06.019.